Day 38 EBIKE101

E-bike Homunculous Diagnosis

Find what is present, what is missing, what is unsafe, and what needs finishing before power or riding.

Tuesday | 2026-09-08

Today’s real job

- Diagnose the unfinished e-bike build without guessing.
- Build a parts and finish-work list.
- Begin safe approved work only after inspection.
- Document every symptom, measurement, and decision.

The diagnosis rule

- Inventory first.
- De-energize before continuity testing.
- Trace source → protection → control → load.
- Power testing comes after mechanical and electrical checks.

System inventory

- Battery present and voltage known?
- Fuse or breaker present and correctly placed?
- Controller mounted and labeled?
- Display, PAS, throttle, brake cutoffs identified?
- Motor cable protected from pull, rub, and wheel movement?

Likely failure points

- Blown fuse or bad fuse holder.
- Loose or mismatched connector.
- Missing throttle or control input.
- Brake cutoff stuck active.
- PAS sensor misaligned.
- Battery switch, BMS, or connector not passing power.

Safety boundary

- No ride testing until instructor clears it.
- No live continuity tests.
- No bypassing fuses.
- No “larger fuse fixes it” nonsense.
- No hands near moving chain, wheel, or rotor during powered tests.

Team roles

- Lead diagnostician: keeps the fault tree organized.
- Meter lead: takes and calls out measurements.
- Recorder: logs symptoms, measurements, changes, and next step.
- Safety spotter: watches hands, wheels, cables, and battery state.

Fault tree order

- 1. Mechanical safe to inspect?
- 2. Battery and protection path verified?
- 3. Controller powered?
- 4. Display wakes?
- 5. Inputs identified?
- 6. Motor phases/hall/PAS/brake signals checked?
- 7. Safe low-power test approved?

Finish-work list

- Must finish before power.
- Must finish before ride.
- Nice to improve if time allows.
- Parts/tools needed.
- Final verification evidence needed.

The lesson underneath

- The smallest cheapest part can stop the largest expensive system.
- A hidden bad fuse can look like a dead motor, controller, or battery.
- Good technicians verify before replacing expensive parts.
- Documentation turns chaos into progress.

Exit target

- By the end of class, every team should know:
- What works.
- What does not work yet.
- What is missing.
- What must be verified before power or riding.
- What the next safe work step is.